DATA SCIENCE MINI PROJECT

Bike Rental Demand Forecasting & Model Interpretability

A time-series forecasting study using SARIMAX and SHAP-based feature analysis

Project component	Implementation
Historical dataset	UCI Bike Sharing Dataset — daily records (day.csv)
Forecasting model	SARIMAX with weekly seasonal differencing (period = 7)
Interpretability	SHAP values for a supplementary Random Forest model
Evaluation window	Final 60 daily observations: 2 November–31 December 2012
Primary tools	Python, pandas, statsmodels, scikit-learn, SHAP, Matplotlib, Seaborn

Executive Summary

This project investigates daily bike rental demand and develops a statistical forecasting baseline. The workflow covers data inspection, time-series stationarity testing, seasonal differencing, SARIMAX modelling, forecast uncertainty, benchmark comparison, and model interpretability.

The selected SARIMAX model achieved an MAE of 1,272.19 and an RMSE of 1,822.18 on the 60-day holdout period. It outperformed a seasonal-naive benchmark on both metrics, although forecast errors remain material. A separate Random Forest model was interpreted with SHAP to estimate the relative contribution of calendar and weather variables.

Important scope note: SHAP explains the supplementary Random Forest model, not the SARIMAX model itself. Feature importance describes model behaviour and should not be interpreted as proof of causation.

Prepared as a reproducible data science project report.

1. Project Overview & Dataset

1.1 Objective

The objective is to model historical sequential business demand, generate forecasts with confidence intervals, and investigate which observed variables are most influential in a complementary predictive model. The project follows a time-aware workflow so that later observations are not used to train the forecasting model.

1.2 Dataset

The analysis uses the daily file (day.csv) from the UCI Bike Sharing Dataset. It contains 731 daily observations spanning 1 January 2011 through 31 December 2012, with 16 original columns. The target variable is **cnt**, representing the total number of bike rentals per day.

Data check	Observed result
Rows × columns	731 × 16
Date range	2011-01-01 to 2012-12-31
Missing values	0
Duplicate rows	0
Mean daily rentals	4,504.35
Minimum / maximum	22 / 8,714

1.3 Variables used

The time-series target is **cnt**. The supplementary Random Forest uses ten explanatory variables: season, yr, mnth, holiday, weekday, workingday, weathersit, temp, hum, and windspeed. The variables casual and registered were intentionally excluded because their sum directly forms cnt, which would introduce target leakage. atemp was also omitted to avoid retaining a highly correlated temperature measure alongside temp.

1.4 Exploratory observations

Daily rental demand shows an overall upward pattern and visible seasonal variation. Monthly averages differ substantially across the observation period. The highest monthly average reported in the exploration was September 2012 (7,285.77), while the lowest was January 2011 (1,231.90). These patterns support the need to examine both trend and seasonality before fitting a forecasting model.

2. Time-Series Preparation & SARIMAX

2.1 Chronological preparation

The date column was converted to datetime, records were sorted chronologically, and the date was set as the index. The final 60 observations were reserved for testing, while the first 671 observations were used for model training. This split preserves temporal order and avoids random shuffling.

Partition	Observations	Date range
Training	671	2011-01-01 to 2012-11-01
Testing	60	2012-11-02 to 2012-12-31

2.2 Stationarity testing

The Augmented Dickey-Fuller (ADF) test was applied to the original cnt series. The p-value was 0.3427, so the null hypothesis of a unit root could not be rejected at conventional significance levels. The original series was therefore treated as non-stationary for this modelling workflow.

Weekly seasonal differencing was then applied using a lag of 7 days: the differenced value is cnt(t) - cnt(t-7). The ADF test on the differenced series returned a statistic of -7.9573 and a p-value of approximately **3.01e-12**, providing strong evidence of stationarity after seasonal differencing.

2.3 Forecasting model

A SARIMAX model with order (1, 0, 1) and seasonal order (1, 1, 1, 7) was fitted to the training series. The model includes a weekly seasonal component and uses seasonal differencing at period 7. The fitted model reported an AIC of 10,849.12. Forecasts were generated for the 60-day test horizon, together with 95% confidence intervals.

The confidence intervals communicate forecast uncertainty; they are not guarantees that the realized values will fall within the interval. Their width in this run indicates considerable uncertainty over parts of the forecast horizon.

2.4 Residual diagnostics

Ljung-Box tests on the fitted residuals returned small p-values at lags 7, 14, and 21 (approximately **3.44e-10**, **8.78e-8**, and **5.45e-6**). This indicates that residual autocorrelation remains. The model therefore does not capture all temporal structure, and its forecasts should be treated as a baseline rather than a fully adequate final model.

3. Forecast Evaluation & Benchmarking

3.1 Evaluation metrics

The forecast was evaluated on the held-out 60 days using MAE, RMSE, MAPE, WAPE, and sMAPE. MAE measures average absolute error in rental-count units. RMSE penalizes larger errors more strongly. Percentage metrics express error relative to actual demand, but MAPE can become unstable when actual values are small.

Model	MAE	RMSE	MAPE	WAPE	sMAPE
SARIMAX	1,272.19	1,822.18	70.56%	28.23%	31.83%
Seasonal Naive	1,488.60	1,977.32	N/A	N/A	N/A

3.2 Interpretation

SARIMAX achieved lower MAE and RMSE than the seasonal-naive benchmark. The MAE improvement was about 14.5%, and the RMSE improvement was about 7.9%, based on the reported values. This suggests the fitted model extracted useful structure beyond simply repeating the demand from seven days earlier.

However, the MAPE value is high. MAPE is especially sensitive to low actual values, which can inflate the percentage error even when absolute errors are comparable. WAPE and sMAPE provide additional perspectives, but they also show that the forecast is not sufficiently accurate to be presented as a high-precision operational forecast.

3.3 Model selection note

A second SARIMAX specification with a linear trend was tested but performed worse on the holdout set (MAE 1,563.35; RMSE 2,179.66; MAPE 85.15%) and produced a convergence warning. The original SARIMAX specification was therefore retained as the best among the tested alternatives, not claimed to be globally optimal.

3.4 Limitations

The dataset covers only two calendar years, limiting the amount of repeated annual seasonality available for learning. Remaining residual autocorrelation suggests that the model is missing temporal information. Further work could test alternative seasonal periods, richer SARIMAX orders, exogenous regressors, transformations, and rolling-origin validation rather than relying on a single holdout split.

4. Model Interpretability with SHAP

4.1 Interpretability design

A supplementary Random Forest Regressor was trained on the same chronological training window and evaluated on the final 60 observations. SHAP TreeExplainer was used to calculate feature-level contributions for the Random Forest predictions. The mean absolute SHAP value summarizes the average magnitude of each feature's contribution across the explained observations.

This is a model-interpretability layer alongside the SARIMAX analysis. It is not a direct decomposition of the SARIMAX forecast. The reported percentages normalize mean absolute SHAP values so that the feature contributions sum to 100% for this analysis.

Feature	Mean |SHAP|	Importance
temp	1,629.19	45.04%
yr	907.19	25.08%
season	340.30	9.41%
hum	322.38	8.91%
mnth	146.31	4.05%
windspeed	107.02	2.96%
weekday	78.55	2.17%
weathersit	62.80	1.74%
workingday	18.24	0.50%
holiday	4.84	0.13%

4.2 Main findings

Temperature is the strongest feature in the supplementary model, accounting for 45.04% of normalized mean absolute SHAP importance. The year indicator follows at 25.08%, while season and humidity contribute 9.41% and 8.91%, respectively. The remaining calendar and weather features have smaller average absolute contributions in this fitted model.

4.3 Reading the SHAP summary plot

The SHAP summary plot displays both the magnitude and direction of feature contributions for individual observations. Each point represents one observation; horizontal position indicates whether that feature pushes the model prediction higher or lower relative to the model's baseline. Point colour typically represents the feature value, from low to high. The plot helps reveal patterns that a ranked bar chart alone cannot show.

4.4 Interpretation cautions

SHAP values explain how this particular Random Forest uses the supplied inputs; they do not establish causal effects. Correlated predictors can share or redistribute importance. The year and month features may also encode broader time trends that are not guaranteed to persist. The findings should therefore be read as predictive associations within this dataset and model, not universal rules about bike demand.

5. Conclusion & Reproducibility

5.1 Conclusion

This project implemented an end-to-end time-series workflow for daily bike rental demand. The original series failed the ADF stationarity test, while weekly seasonal differencing produced a stationary differenced series according to the test. A SARIMAX (1,0,1) × (1,1,1,7) model generated a 60-day forecast and 95% confidence intervals.

On the held-out period, SARIMAX outperformed the seasonal-naive benchmark on MAE and RMSE, but the remaining error and residual autocorrelation show that the model is a useful baseline rather than a production-ready forecasting solution. The supplementary SHAP analysis identified temperature and year as the two largest contributors to Random Forest predictions.

5.2 Exported outputs

File	Contents
outputs/forecast_results.csv	Date, actual demand, predicted demand, lower and upper confidence limits
outputs/model_metrics.csv	SARIMAX and seasonal-naive evaluation metrics
outputs/shap_feature_importance.csv	Mean absolute SHAP values and normalized importance percentages

5.3 Reproducibility and suggested next steps

The notebook should retain the chronological split, model configuration, feature list, random seed, and export steps so the analysis can be reproduced. Recommended extensions include rolling-origin cross-validation, residual modelling, comparison with Prophet or alternative SARIMA configurations, and adding carefully selected exogenous variables. Any revised model should be compared against the same baseline and evaluation window.

5.4 Dataset attribution

Dataset: UCI Bike Sharing Dataset, available from the UCI Machine Learning Repository:
https://archive.ics.uci.edu/dataset/275/bike+sharing+dataset
The dataset is attributed to the original dataset authors and repository. Confirm and retain the repository's stated license and citation requirements when redistributing the raw data with the project.

End of report